

# A single kinetic defect in a run-and-tumble ring: exact spectrum, exceptional points, and relaxation crossover

**Abstract.** We solve the spectral problem for one continuous-time run-and-tumble particle on an  $n$ -site ring when the tumbling rate at one site is  $\omega_0$  and the bulk rate is  $\omega$ . The defect changes the  $2n \times 2n$  generator by rank one. The characteristic polynomial is the homogeneous polynomial plus  $(\omega_0 - \omega)/n$  times its  $\omega$ -derivative. This identity gives Chebyshev and transfer-matrix quantization formulas, a parity factorization, and finite polynomial tests for every Jordan block. We separately resolve the singular flat-band point  $\omega = 1$ , where the eigenvalue  $-2$  can carry Jordan blocks of sizes two or three. The infinite-line Green function gives an exact criterion for exponentially localized defect modes. For fixed positive bulk rate, reflection protects one homogeneous long-wave mode; consequently a slow defect, including  $\omega_0 \asymp n^{-1}$ , does not produce a new spectral-gap scaling. The nontrivial  $n^{-1}$  crossover occurs when the bulk rate itself is of order  $n^{-1}$ . Finally, the standard telegraph equation misses the lattice Poisson-diffusion term at fixed  $\omega$ , while a diffusion-corrected telegraph equation agrees through order  $n^{-2}$ .

**Keywords.** Run-and-tumble particle; kinetic defect; non-Hermitian spectrum; exceptional point; localized mode; spectral gap; telegraph equation.

## 1. Introduction

Persistent random walks provide a microscopic description of motion with a finite directional memory. Continuum formulations with space-dependent turning statistics arise, for example, in the theory of chemotactic random walks [Sch93], while lattice run-and-tumble models retain the individual hopping and reversal events and connect directly to nonequilibrium particle systems [TTCB11]. For a homogeneous ring, the one-particle spectrum, its exceptional points, and the associated relaxation transition are known explicitly [MBE19]. Spatial inhomogeneity has been studied through position-dependent speed [AG19], quenched disorder in speed and tumbling [BWK<sup>+</sup>19, Secs. 2.1.2–2.1.3 and 2.2.2–2.2.3], and saturating space-dependent rates on the line [JC24, Sec. II, Eqs. (3)–(4)]. Those works emphasize probability laws, steady states, or relaxation in extended media. The present problem instead asks for the full finite-ring characteristic polynomial and Jordan structure of a single lattice tumbling defect, together with its large-ring spectral ordering.

This paper asks how that spectral picture changes when the tumbling rate is altered at one site. The inhomogeneity is local but the generator is non-normal, so eigenvalue crossings, Jordan structure, and long-wave relaxation must be tracked together. The central observation is that the defect is rank one in the polarization channel. It leads to a finite characteristic identity, a parity decomposition, an infinite-line Green-function criterion including a compact defect mode, and a direct comparison between lattice and telegraph relaxation. The fixed-rate and bulk-critical limits separate the effects of a local defect from those of a vanishing bulk tumbling rate.

## 2. Model and notation

**Definition 2.1** (Defective run-and-tumble generator). Let  $n \geq 2$ , let  $\omega > 0$ , and let  $\omega_0 \geq 0$ . Sites are  $j \in \mathbb{Z}/n\mathbb{Z}$ , orientations are  $\sigma \in \{+, -\}$ , and

$$\omega_j = \begin{cases} \omega_0, & j = 0, \\ \omega, & j \neq 0. \end{cases}$$

The forward generator  $L = L_n(\omega, \omega_0)$  is

$$(Lu)_j^+ = u_{j-1}^+ - (1 + \omega_j)u_j^+ + \omega_j u_j^-, \quad (2.1)$$

$$(Lu)_j^- = u_{j+1}^- - (1 + \omega_j)u_j^- + \omega_j u_j^+. \quad (2.2)$$

Thus hopping in the current orientation has rate one and tumbling has rate  $\omega_j$ . Put

$$\delta = \omega_0 - \omega, \quad q_k = \frac{2\pi k}{n}, \quad c_k = \cos q_k, \quad s_k = \sin q_k.$$

Equations (2.1)–(2.2) use the same one-particle convention as equations (6)–(10) of [MBE19]. That paper records the homogeneous eigenvalues in its equation (10), the homogeneous exceptional points in equation (13), and the first-mode relaxation rate in equations (20)–(21).

**Lemma 2.2** (Rank-one defect). Let  $L^{(0)} = L_n(\omega, \omega)$ . If  $d = e_{0,+} - e_{0,-}$ , then

$$L = L^{(0)} - \delta dd^\top.$$

In particular, the perturbation has rank one when  $\delta \neq 0$ .

*Proof.* At site zero, changing  $\omega$  to  $\omega_0$  adds

$$\delta \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} = -\delta \begin{pmatrix} 1 \\ -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \end{pmatrix}.$$

Every other matrix entry is unchanged. ■

### 3. Three exact characteristic equations

**Definition 3.1** (Homogeneous factors). For  $\lambda \in \mathbb{C}$ , set

$$a = \lambda + 1 + \omega, \quad (3.1)$$

$$x = \frac{a^2 + 1 - \omega^2}{2a}, \quad (3.2)$$

$$D_k(\lambda) = (\lambda + 1 - c_k)(\lambda + 1 + 2\omega - c_k) + s_k^2 = a^2 + 1 - \omega^2 - 2ac_k. \quad (3.3)$$

Although  $x$  is undefined at  $a = 0$ , each expression below containing  $a^n T_n(x)$  is understood by its unique polynomial continuation.

**Lemma 3.2** (*Homogeneous polynomial*). The homogeneous characteristic polynomial is

$$\begin{aligned} P_n^{(0)}(\lambda; \omega) &= \det(\lambda I - L^{(0)}) \\ &= \prod_{k=0}^{n-1} D_k(\lambda) \end{aligned} \quad (3.4)$$

$$= 2a^n (T_n(x) - 1), \quad (3.5)$$

where  $T_n$  is the Chebyshev polynomial of the first kind. The two roots of  $D_k$  are

$$\lambda_k^\pm = c_k - 1 - \omega \pm \sqrt{\omega^2 - s_k^2}.$$

*Proof.* Fourier transformation gives the block

$$W_k = \begin{pmatrix} e^{iq_k} - 1 - \omega & \omega \\ \omega & e^{-iq_k} - 1 - \omega \end{pmatrix}.$$

Its characteristic polynomial is  $D_k$ , which proves (3.4) and the displayed roots. Since  $D_k = 2a(x - c_k)$  and

$$\prod_{k=0}^{n-1} (x - c_k) = 2^{1-n} (T_n(x) - 1),$$

equation (3.5) follows. ■

**Theorem 3.3** (*Exact finite characteristic polynomial*). For every  $n \geq 2$ ,  $\omega > 0$ , and  $\omega_0 \geq 0$ ,

$$P_n(\lambda; \omega, \omega_0) := \det(\lambda I - L_n(\omega, \omega_0)) \quad (3.6)$$

$$= P_n^{(0)}(\lambda; \omega) + \frac{\delta}{n} \partial_\omega P_n^{(0)}(\lambda; \omega) \quad (3.7)$$

$$= P_n^{(0)}(\lambda; \omega) \left[ 1 + \frac{2\delta}{n} \sum_{k=0}^{n-1} \frac{\lambda + 1 - c_k}{D_k(\lambda)} \right]. \quad (3.8)$$

Equation (3.8) is interpreted by polynomial continuation at a homogeneous eigenvalue. Equivalently,

$$P_n = 2a^n (T_n(x) - 1) + 2\delta a^{n-1} (T_n(x) - 1) + \delta a^{n-2} \lambda(\lambda + 2) U_{n-1}(x), \quad (3.9)$$

where  $U_{n-1}$  is the Chebyshev polynomial of the second kind and the right-hand side again denotes its polynomial continuation.

*Proof.* Let  $B = \lambda I - L^{(0)}$ . Lemma 2.2 and the matrix determinant lemma give

$$\det(\lambda I - L) = \det B(1 + \delta d^\top B^{-1} d).$$

In Fourier space,

$$d^\top B^{-1} d = \frac{1}{n} \sum_{k=0}^{n-1} \frac{2(\lambda + 1 - c_k)}{D_k(\lambda)},$$

which proves (3.8). Direct differentiation of (3.4) gives

$$\partial_\omega P_n^{(0)} = P_n^{(0)} \sum_k \frac{2(\lambda + 1 - c_k)}{D_k},$$

and hence (3.7). Finally,

$$\partial_\omega x = \frac{\lambda(\lambda + 2)}{2a^2}, \quad T'_n(x) = nU_{n-1}(x),$$

and differentiating (3.5) proves (3.9). ■

**Theorem 3.4** (*Transfer-matrix quantization*). For a local rate  $r \geq 0$ , define

$$T_r(\lambda) = \frac{1}{\lambda + 1 + r} \begin{pmatrix} 1 & r \\ -r & (\lambda + 1 + r)^2 - r^2 \end{pmatrix}.$$

Then  $\det T_r = 1$ , and, away from the displayed denominators,

$$P_n(\lambda; \omega, \omega_0) = a^{n-1}(\lambda + 1 + \omega_0) \{ \text{tr}[T_\omega(\lambda)^{n-1} T_{\omega_0}(\lambda)] - 2 \}.$$

After multiplication by the denominators, this identity holds for every  $\lambda$  by polynomial continuation. When  $a \neq 0$  and  $\lambda + 1 + \omega_0 \neq 0$ , an eigenvalue is equivalently characterized by

$$\text{tr}[T_\omega(\lambda)^{n-1} T_{\omega_0}(\lambda)] = 2.$$

At a zero of either denominator, the cleared polynomial identity (3.4), rather than the undefined trace in (3.4), is the eigenvalue criterion. Moreover, if  $t = \text{tr} T_\omega = 2x$ , then

$$T_\omega^{n-1} = U_{n-2}(x)T_\omega - U_{n-3}(x)I,$$

with  $U_{-1} = 0$ .

*Proof.* The eigenvalue equations at site  $j$  are

$$\begin{aligned} (\lambda + 1 + \omega_j)u_j^+ - \omega_j u_j^- &= u_{j-1}^+, \\ (\lambda + 1 + \omega_j)u_j^- - \omega_j u_j^+ &= u_{j+1}^-. \end{aligned}$$

Consequently

$$\begin{pmatrix} u_j^+ \\ u_{j+1}^- \end{pmatrix} = T_{\omega_j}(\lambda) \begin{pmatrix} u_{j-1}^+ \\ u_j^- \end{pmatrix}.$$

Periodic boundary conditions require the product monodromy to have eigenvalue one. Its determinant is one, so this is equivalent to trace two. To identify the polynomial, retain temporarily arbitrary rates  $r_0, \dots, r_{n-1}$ , denote the resulting generator by  $L_n(r_0, \dots, r_{n-1})$ , and put  $b_j = \lambda + 1 + r_j$ . Gaussian elimination of the  $2n$  eigenvalue equations, in the order  $u_0^+, u_1^+, \dots, u_{n-1}^+$ , has pivots  $b_0, \dots, b_{n-1}$ . If  $M = T_{r_{n-1}} \cdots T_{r_0}$ , the two remaining cyclic boundary equations have coefficient matrix  $M - I$ . The cyclic row ordering contributes one minus sign, and  $-\det(M - I) = \text{tr } M - 2$  because  $\det M = 1$ . Consequently

$$\det(\lambda I - L_n(r_0, \dots, r_{n-1})) = \left( \prod_{j=0}^{n-1} b_j \right) [\text{tr}(T_{r_{n-1}} \cdots T_{r_0}) - 2]. \quad (3.10)$$

This is an identity of rational functions where all  $b_j \neq 0$ ; multiplication by the displayed pivots makes both sides polynomials, so the identity extends to every  $\lambda$ . Taking  $r_0 = \omega_0$  and all other rates equal to  $\omega$ , and using cyclicity of the trace, gives (3.4), including algebraic multiplicities. Equation (3.4) is the Cayley–Hamilton identity for a unimodular  $2 \times 2$  matrix. ■

#### 4. Parity factorization and every finite exceptional point

**Definition 4.1** (Kinetic reflection). Define the involution

$$(Ru)_j^+ = u_{-j}^-, \quad (Ru)_j^- = u_{-j}^+.$$

Write  $\mathcal{H}_\pm = \ker(R \mp I)$ .

**Lemma 4.2** (*Protected sector*). *The generator commutes with  $R$ , the defect vector satisfies  $Rd = -d$ , and*

$$L|_{\mathcal{H}_+} = L^{(0)}|_{\mathcal{H}_+}.$$

*Thus the defect changes only the reflection-odd sector.*

*Proof.* Reflection reverses position and orientation simultaneously, leaving the directed hopping rules and the rate field  $\omega_j$  invariant. At the fixed site zero it exchanges the two entries of  $d$ , hence  $Rd = -d$ . Lemma 2.2 then vanishes on  $\mathcal{H}_+$ . ■

**Definition 4.3** (Parity polynomials). Put  $m = \lfloor (n-1)/2 \rfloor$  and let  $\varepsilon_n$  be one for even  $n$

and zero for odd  $n$ . Define

$$H_{n,+}(\lambda) = \lambda(\lambda + 2)^{\varepsilon_n} \prod_{k=1}^m D_k(\lambda), \quad (4.1)$$

$$H_{n,-}^{(0)}(\lambda) = (\lambda + 2\omega)(\lambda + 2 + 2\omega)^{\varepsilon_n} \prod_{k=1}^m D_k(\lambda), \quad (4.2)$$

$$G_n(\lambda) = \frac{1}{n} \left[ \frac{1}{\lambda + 2\omega} + \frac{\varepsilon_n}{\lambda + 2 + 2\omega} + 2 \sum_{k=1}^m \frac{\lambda + 1 - c_k}{D_k(\lambda)} \right], \quad (4.3)$$

$$Q_{n,-}(\lambda) = H_{n,-}^{(0)}(\lambda)(1 + 2\delta G_n(\lambda)). \quad (4.4)$$

Every quotient in (4.4) cancels, so  $Q_{n,-}$  is a monic polynomial of degree  $n$ . An entirely polynomial version is

$$Q_{n,-} = H_{n,-}^{(0)} + \frac{2\delta}{n} \left[ \frac{H_{n,-}^{(0)}}{\lambda + 2\omega} + \varepsilon_n \frac{H_{n,-}^{(0)}}{\lambda + 2 + 2\omega} + 2 \sum_{k=1}^m (\lambda + 1 - c_k) \frac{H_{n,-}^{(0)}}{D_k} \right]. \quad (4.5)$$

**Theorem 4.4** (*Exact parity factorization*). *The characteristic polynomial factors as*

$$P_n(\lambda; \omega, \omega_0) = H_{n,+}(\lambda) Q_{n,-}(\lambda).$$

*The first factor is the characteristic polynomial on  $\mathcal{H}_+$ ; the second is the characteristic polynomial on  $\mathcal{H}_-$ .*

*Proof.* The modes  $q_k$  and  $-q_k$  form a four-dimensional Fourier subspace, which splits into two isomorphic two-dimensional  $R$ -eigenspaces. Each has characteristic polynomial  $D_k$ . At  $q = 0$ , the vectors  $(1, 1)$  and  $(1, -1)$  have parities  $+$  and  $-$  and eigenvalues  $0$  and  $-2\omega$ . When  $n$  is even, the  $q = \pi$  density and polarization vectors have parities  $+$  and  $-$  and eigenvalues  $-2$  and  $-2 - 2\omega$ . This gives (4.1)–(4.2). Lemma 4.2 and the rank-one determinant lemma restricted to  $\mathcal{H}_-$  give (4.4). Multiplication proves (4.4). ■

**Definition 4.5** (Exceptional point). For an eigenvalue  $\mu$ , let  $m_{\text{alg}}(\mu)$  be its algebraic multiplicity and  $m_{\text{geo}}(\mu) = \dim \ker(\mu I - L)$  its geometric multiplicity. We call  $(\omega, \omega_0, \mu)$  an exceptional point when  $m_{\text{alg}}(\mu) > m_{\text{geo}}(\mu)$ . Write  $J_r(\mu)$  for a Jordan block of size  $r$  at  $\mu$ .

**Lemma 4.6** (*No inter-wave coincidences away from the flat band*). *If  $c_k \neq c_\ell$  and  $D_k(\mu) = D_\ell(\mu) = 0$ , then  $\omega = 1$  and  $\mu = -2$ .*

*Proof.* Subtracting the two equations gives  $2a(c_k - c_\ell) = 0$ , hence  $a = 0$ . Substitution into (3.3) gives  $1 - \omega^2 = 0$ . Since  $\omega > 0$ , this means  $\omega = 1$  and then  $\mu = -1 - \omega = -2$ . ■

**Theorem 4.7** (*Jordan classification for  $\omega \neq 1$* ). *Assume  $\omega \neq 1$ .*

1. In the protected sector, the endpoint factors in (4.1) give one-dimensional blocks. For each  $1 \leq k \leq m$ , a root of  $D_k$  gives one  $J_1$ , except that

$$\omega = |s_k|, \quad \mu = c_k - 1 - \omega. \quad (4.6)$$

gives one  $J_2(\mu)$ .

2. In the affected sector, if

$$Q_{n,-}(\mu) = Q'_{n,-}(\mu) = \cdots = Q_{n,-}^{(r-1)}(\mu) = 0, \quad Q_{n,-}^{(r)}(\mu) \neq 0, \quad (4.7)$$

then the complete affected-sector contribution at  $\mu$  is one block  $J_r(\mu)$ .

3. The full Jordan form at  $\mu$  is the direct sum of the blocks in items 1 and 2. In particular, a crossing between the two parity sectors is semisimple unless one of its sector blocks has size greater than one.

Consequently, (4.6) together with (4.7) is an exact finite criterion for every exceptional point when  $\omega \neq 1$ . The order  $r$  in (4.7) classifies exceptional points of arbitrary order.

*Proof.* The discriminant of  $D_k$  is  $4(\omega^2 - s_k^2)$ . At a zero discriminant its Fourier block is not scalar and hence is a single  $J_2$ ; otherwise each root is simple. Lemma 4.6 excludes coincidences between distinct  $c_k$  values.

It remains to show that the affected restriction is cyclic. In each two-dimensional homogeneous parity block, the spectral weight of the defect vector has numerator  $2(\lambda + 1 - c_k)$ . This numerator and  $D_k$  have no common root for an interior wave number: a common root would give  $\lambda + 1 = c_k$ , after which  $D_k = s_k^2 \neq 0$ . The endpoint weights in (4.3) are also nonzero. Lemma 4.6 makes the block minimal polynomials coprime. The defect vector is therefore cyclic for the homogeneous odd restriction. If  $A$  has cyclic vector  $d$ , then  $d$  is also cyclic for  $A - \delta dd^\top$ , because each  $(A - \delta dd^\top)^j d$  equals  $A^j d$  plus a linear combination of lower powers. Hence the minimal and characteristic polynomials of the affected restriction both equal  $Q_{n,-}$ . A cyclic matrix has one Jordan block for each eigenvalue, of size equal to the root order. The direct-sum assertion follows from the invariant parity splitting. ■

**Theorem 4.8** (*The complete flat-band classification at  $\omega = 1$* ). Let  $\omega = 1$  and put  $\mu = -2$ .

1. If  $\omega_0 = 1$  and  $4 \nmid n$ , the contribution at  $\mu$  is  $nJ_1(\mu)$ .
2. If  $\omega_0 = 1$  and  $4 \mid n$ , the contribution is

$$2J_2(\mu) \oplus (n-2)J_1(\mu).$$

3. If  $\omega_0 \neq 1$  and  $n \equiv 1$  or  $2 \pmod{4}$ , the contribution is  $(n-1)J_1(\mu)$ .
4. If  $\omega_0 \neq 1$  and  $n \equiv 0 \pmod{4}$ , the contribution is

$$J_2(\mu) \oplus (n-2)J_1(\mu).$$

5. If  $n \equiv 3 \pmod{4}$ ,  $\omega_0 \neq 1$ , and  $\omega_0 \neq (n-1)/(n+1)$ , the contribution is again  $J_2(\mu) \oplus (n-2)J_1(\mu)$ .

6. If  $n \equiv 3 \pmod{4}$  and  $\omega_0 = (n-1)/(n+1)$ , the contribution is

$$J_3(\mu) \oplus (n-2)J_1(\mu).$$

For eigenvalues  $\lambda \neq -2$ , every protected-sector root is a one-dimensional block. Remove the exact power of  $(\lambda+2)$  specified above from  $Q_{n,-}$ . Every root of the remaining affected polynomial contributes one Jordan block, so the derivative test (4.7) applied to that reduced polynomial classifies every remaining affected block at  $\omega = 1$ . The full Jordan form is the direct sum of the protected and affected blocks.

*Proof.* Write  $t = \lambda + 2$ . At  $\omega = 1$ ,

$$D_k = t(t - 2c_k), \quad P_n = t^{n-1}F_n(t),$$

where, for  $\delta = \omega_0 - 1$ ,

$$F_n(t) = 2t(T_n(t/2) - 1) + \delta [2(T_n(t/2) - 1) + (t-2)U_{n-1}(t/2)]. \quad (4.8)$$

The values

$$T_n(0) = \cos(n\pi/2), \quad U_{n-1}(0) = \sin(n\pi/2)$$

give the following algebraic multiplicities. For  $\delta = 0$  they are  $n$ , except that  $4 \mid n$  gives  $n+2$ . For  $\delta \neq 0$  they are  $n-1$  when  $n \equiv 1, 2 \pmod{4}$ , and  $n$  when  $n \equiv 0, 3 \pmod{4}$ . In the last congruence class,  $F'_n(0) = -2 - \delta(n+1)$ ; it vanishes exactly at  $\delta = -2/(n+1)$ . For  $n \equiv 3 \pmod{4}$  one has  $U''_{n-1}(0) = n^2 - 1$ , and direct differentiation of (4.8) at the tuned value gives

$$F''_n(0) = -2n + \frac{U''_{n-1}(0)}{n+1} = -(n+1) \neq 0. \quad (4.9)$$

Thus the tuned multiplicity is  $n+1$ .

The bulk equations at  $t = 0$  read

$$u_j^- = -u_{j-1}^+ \quad (j \neq 0).$$

Together they impose this relation for every  $j$ . The defect equation adds no constraint when  $\omega_0 = 1$  and otherwise adds the single constraint  $u_0^+ + u_{n-1}^+ = 0$ . Thus the geometric multiplicity is  $n$  in the homogeneous case and  $n-1$  otherwise. Fourier decomposition shows that the two excess algebraic dimensions when  $4 \mid n$  and  $\delta = 0$  are the two  $q = \pm\pi/2$  blocks, each a  $J_2$ . For  $\delta \neq 0$ , the parity factorization places the  $4 \mid n$  excess in one protected  $J_2$ .

It remains to distinguish one  $J_3$  from two  $J_2$  blocks in the tuned  $n \equiv 3 \pmod{4}$  case. Put  $N_0 = L_n(1, 1) + 2I$  and  $N = L_n(1, \omega_0) + 2I = N_0 - \delta dd^\top$ . Because  $4 \nmid n$ , zero is a semisimple eigenvalue of  $N_0$  with eigenspace  $E = \ker N_0$  of dimension  $n$ . Let  $\Pi$  be its spectral projection. Write

$$\det(tI - N_0) = t^n h(t), \quad h(0) \neq 0.$$



Semisimplicity gives  $(tI - N_0)^{-1} = t^{-1}\Pi + O(1)$ . The rank-one determinant formula then shows that the coefficient of  $t^{n-1}$  in  $\det(tI - N)$  is  $\delta h(0)d^\top \Pi d$ . The multiplicity computation above shows that this coefficient vanishes for  $n \equiv 3 \pmod{4}$ , and therefore

$$d^\top \Pi d = 0. \quad (4.10)$$

The eigenspace of  $N$  is  $K = E \cap \ker d^\top$ , which has dimension  $n - 1$ . If  $v \in K \cap \text{ran } N$ , write  $v = Nu$  and apply  $\Pi$ :

$$v = \Pi Nu = -\delta \Pi d(d^\top u). \quad (4.11)$$

Thus  $K \cap \text{ran } N$  has dimension at most one. It has dimension one because the algebraic multiplicity exceeds the geometric multiplicity. Hence exactly one Jordan block at zero for  $N$  is nontrivial. It is a  $J_2$  when the algebraic multiplicity is  $n$ , and the additional algebraic order at  $\delta = -2/(n+1)$  makes that same block a  $J_3$ . This proves all six cases and excludes the alternative of two  $J_2$  blocks in the tuned case.

For completeness, put  $A_-^{(0)} = L^{(0)}|_{\mathcal{H}_-}$  and consider an affected eigenvalue  $\lambda \neq -2$ . If  $\delta \neq 0$ , it cannot equal a nonflat homogeneous root: the corresponding residue in (4.3) is nonzero, so  $Q_{n,-}$  does not vanish at that pole. Hence  $\lambda I - A_-^{(0)}$  is invertible. The rank-one eigenvalue equation makes its eigenspace the one-dimensional span of  $(\lambda I - A_-^{(0)})^{-1}d$ . Its root order therefore gives the size of its unique Jordan block. If  $\delta = 0$ , all nonflat roots are simple by Lemma 4.6. This proves the final classification without asserting cyclicity at the removed flat primary component. ■

**Corollary 4.9** (*Finite algebraic exceptional set*). *For fixed  $n$ , all non-flat affected exceptional parameters lie on the explicit algebraic set*

$$\text{Res}_\lambda(Q_{n,-}, \partial_\lambda Q_{n,-}) = 0,$$

with  $Q_{n,-}$  given by (4.5). At  $\omega = 1$ , the flat factor is removed before taking the resultant. Root orders and theorems 4.7–4.8 distinguish exceptional points from semisimple parity crossings without a numerical tolerance.

*Proof.* The resultant vanishes exactly when the two displayed polynomials share a root. The preceding two theorems identify the Jordan block attached to every such root and separately account for the flat factor. ■

## 5. Defect-localized modes

**Definition 5.1** (Bulk spectral set). For the infinite translation-invariant chain, define

$$\Sigma_\omega = \left\{ c - 1 - \omega \pm \sqrt{\omega^2 - (1 - c^2)} : c \in [-1, 1] \right\}.$$

For  $\lambda \notin \Sigma_\omega$  with  $a \neq 0$ , retain  $a$  and  $x$  from (3.1)–(3.2), and choose

$$s(\lambda) = \sqrt{x(\lambda)^2 - 1}$$

as the standard branch on  $x \in \mathbb{C} \setminus [-1, 1]$  satisfying  $s/x \rightarrow 1$  as  $|x| \rightarrow \infty$ . Equivalently, it is the branch satisfying  $s(\lambda) \sim x(\lambda)$  as  $|\lambda| \rightarrow \infty$ . Put  $z = x - s$ , so  $|z| < 1$ .

**Theorem 5.2** (*Infinite-line localization criterion*). Let  $\delta \neq 0$  and  $\lambda \notin \Sigma_\omega$ . First suppose  $a \neq 0$ . Then  $\lambda$  is an exponentially localized eigenvalue of the infinite chain with the same single tumbling defect if and only if

$$1 + 2\delta g_\infty(\lambda) = 0, \quad (5.1)$$

where

$$g_\infty(\lambda) = \frac{1}{2a} \left( 1 + \frac{\lambda + 1 - x}{s} \right) = \frac{z(\lambda + 1 - z)}{a(1 - z^2)}. \quad (5.2)$$

The localization length is

$$\xi(\lambda) = \frac{-1}{\log |z|}.$$

Equivalently, with  $a = \lambda + 1 + \omega$ , the pair  $(a, z)$  satisfies

$$a^2 + 1 - \omega^2 - a(z + z^{-1}) = 0, \quad (5.3)$$

$$a(1 - z^2) + 2\delta z(a - \omega - z) = 0, \quad 0 < |z| < 1. \quad (5.4)$$

After elimination of  $a$ , every admissible  $z$  is a root of

$$\begin{aligned} 0 &= 2\delta z^3 + (2\delta\omega + 1 - \omega^2)z^2 \\ &\quad + (4\delta^2\omega + 4\delta\omega^2 - 2\delta)z + (2\delta\omega + \omega^2 - 1), \end{aligned} \quad (5.5)$$

and, when the denominator is nonzero,

$$a = \frac{2\delta z(\omega + z)}{1 - z^2 + 2\delta z}, \quad \lambda = a - 1 - \omega. \quad (5.6)$$

A root of (5.5) is accepted only when (5.1), or equivalently both (5.3) and (5.4), is satisfied; this removes roots introduced by elimination or by clearing denominators.

There is one additional compact case, at the point where  $a = 0$ . If  $\omega \neq 1$ , then

$$\lambda = -1 - \omega \quad \text{is localized if and only if} \quad \omega_0 = \frac{\omega^2 + 1}{2\omega}. \quad (5.7)$$

Its decay root is  $z = 0$ , its localization length is  $\xi = 0$ , and its eigenvector is supported at the defect and its two neighboring sites. Thus the admissible range in the preceding formulas is  $0 < |z| < 1$ ;  $z = 0$  is accepted precisely in (5.7).

*Proof.* The polarization entry of the bulk resolvent at the defect site is

$$\begin{aligned} g_\infty(\lambda) &= \frac{1}{2\pi} \int_0^{2\pi} \frac{\lambda + 1 - \cos q}{a^2 + 1 - \omega^2 - 2a \cos q} dq \\ &= \frac{1}{2a} \left[ 1 + (\lambda + 1 - x) \frac{1}{2\pi} \int_0^{2\pi} \frac{dq}{x - \cos q} \right] \\ &= \frac{1}{2a} \left( 1 + \frac{\lambda + 1 - x}{s} \right). \end{aligned}$$

The rank-one eigenvalue equation is (5.1). Since  $x = (z + z^{-1})/2$  and  $s = (z^{-1} - z)/2$ , elementary simplification gives the second expression in (5.2), and the Green kernel decays

as  $z^{|j|}$ . Clearing the denominator in (5.1) gives (5.4); the bulk dispersion relation gives (5.3). Eliminating  $a$  gives (5.5), while solving (5.4) for  $a$  gives (5.6).

It remains to evaluate the point omitted by division by  $a$ . At  $\lambda = -1 - \omega$  the Fourier denominator is the nonzero constant  $1 - \omega^2$ , and hence

$$g_\infty(-1 - \omega) = \frac{1}{2\pi} \int_0^{2\pi} \frac{-\omega - \cos q}{1 - \omega^2} dq = \frac{\omega}{\omega^2 - 1}.$$

The secular equation is therefore equivalent to  $\delta = (1 - \omega^2)/(2\omega)$ , which is exactly (5.7). In Fourier variables the associated resolvent vector is proportional to

$$\begin{pmatrix} -\omega - e^{-iq} \\ \omega + e^{iq} \end{pmatrix}.$$

Its inverse Fourier transform is supported at the defect and its two neighbors. This proves the compact case and also shows directly that it is the limiting root  $z = 0$ . ■

**Proposition 5.3** (*Finite-ring convergence of a localized mode*). Suppose (5.1) has a simple root  $\lambda_\infty$  with decay root  $z_\infty$ ,  $0 < |z_\infty| < 1$ . Then, for all sufficiently large  $n$ , the finite polynomial  $Q_{n,-}$  has a unique root  $\lambda_n$  near  $\lambda_\infty$ , and

$$\lambda_n - \lambda_\infty = O(|z_\infty|^n).$$

The corresponding right eigenvector has site mass bounded by  $C|z_\infty|^{2 \operatorname{dist}(j,0)}$  after normalization. In the compact case (5.7), the infinite and finite eigenvalues agree exactly and the eigenvector is supported where  $\operatorname{dist}(j,0) \leq 1$ .

*Proof.* The root-of-unity sum of the bulk Green kernel is the exact identity

$$G_n(\lambda) = \frac{1}{2a} \left[ 1 + \frac{\lambda + 1 - x}{s} \frac{1 + z^n}{1 - z^n} \right]. \quad (5.8)$$

Indeed, Fourier expansion of  $(x - \cos q)^{-1}$  gives

$$\frac{1}{n} \sum_{k=0}^{n-1} \frac{1}{x - \cos q_k} = \frac{1}{s} \frac{1 + z^n}{1 - z^n}.$$

Both sides of (5.8) are meromorphic functions of  $\lambda$ . The identity therefore continues across the infinite-volume cut to every point away from a finite-ring homogeneous pole; the two boundary choices of  $s$  give the same continued function. Consequently

$$G_n - g_\infty = \frac{\lambda + 1 - x}{as} \frac{z^n}{1 - z^n}. \quad (5.9)$$

Choose a closed disk  $B$  about  $\lambda_\infty$  disjoint from  $\Sigma_\omega$  and containing no other zero of  $1 + 2\delta g_\infty$ . Shrinking it if necessary, choose

$$\rho = \max_{\lambda \in B} |z(\lambda)| < r < 1.$$

Equation (5.9) and its  $\lambda$ -derivative are bounded on  $B$  by  $C_B r^n$ : for the derivative, the only additional factor is  $n\rho^{n-1} = O(r^n)$ . Rouché's theorem on  $\partial B$ , followed by the simple-zero estimate at  $\lambda_\infty$ , gives one zero  $\lambda_n$ . Evaluation of (5.9) at  $\lambda_\infty$ , together with  $z(\lambda_n) = z_\infty + O(\lambda_n - \lambda_\infty)$ , gives  $\lambda_n - \lambda_\infty = O(|z_\infty|^n)$ .

The Fourier coefficients of the two bulk resolvent entries are linear combinations of  $z^{|j|}$  and  $z^{|j\pm 1|}$ . Their periodizations are bounded by

$$\frac{r^{\text{dist}(j,0)} + r^{n-\text{dist}(j,0)}}{1 - r^n}.$$

The coefficients converge to their nonzero infinite-volume values, so normalization changes this bound by a fixed factor. Using  $z(\lambda_n) = z_\infty + O(|z_\infty|^n)$  gives the stated site-mass estimate.

In the compact case, the Laurent-polynomial Fourier vector displayed in the proof of Theorem 5.2 is an exact finite-ring eigenvector for every  $n$  (with the two neighbors identified when  $n = 2$ ), proving the last assertion.  $\blacksquare$

## 6. Large-ring spectral gap

**Definition 6.1** (Spectral gap). For  $\omega > 0$  and  $n \geq 2$ , irreducibility makes zero a simple eigenvalue. Define

$$\gamma_n(\omega, \omega_0) = -\max\{\Re \lambda : \lambda \in \text{spec } L_n(\omega, \omega_0), \lambda \neq 0\}.$$

Put  $q = 2\pi/n$  and

$$\gamma_n^{\text{hom}}(\omega) = 1 - \cos q + \omega - \sqrt{\omega^2 - \sin^2 q}, \quad (6.1)$$

where the square root is positive once  $n$  is sufficiently large for fixed  $\omega > 0$ .

**Lemma 6.2** (*Protected first mode and affected pole*). For every  $n$ , the protected sector contains  $\lambda_1^+ = -\gamma_n^{\text{hom}}$ . For fixed  $\omega > 0$  and  $\omega_0 \geq 0$ , the affected root born from the same homogeneous pole is

$$\lambda_{1,-} = \lambda_1^+ + \frac{\delta(1+\omega)}{\omega^2(1+\omega_0)} \frac{q^2}{n} + O(|\delta|n^{-4}). \quad (6.2)$$

*Proof.* The first assertion is Theorem 4.4. Near  $\lambda_1^+$ , the two terms  $k = 1, n-1$  in the full Fourier Green function have residue

$$R_1 = \frac{\lambda_1^+ + 1 - c_1}{\lambda_1^+ - \lambda_1^-} = -\frac{\omega - r_1}{2r_1}, \quad r_1 = \sqrt{\omega^2 - \sin^2 q}.$$

Thus

$$G_n(\lambda) = \frac{2R_1}{n(\lambda - \lambda_1^+)} + G_n^{\text{reg}}(\lambda).$$

At  $\lambda = 0$ , every nonzero-wave summand is exactly  $1/[2(1+\omega)]$ . Splitting the sum into  $|q_k| \leq n^{-1/2}$  and its complement and using  $\lambda_1^+ = O(n^{-2})$  gives

$$G_n^{\text{reg}}(\lambda_1^+) = \frac{1}{2(1+\omega)} + O(n^{-1}), \quad (G_n^{\text{reg}})'(\lambda) = O(n)$$

for  $|\lambda - \lambda_1^+| = O(n^{-3})$ . Solving  $1 + 2\delta G_n = 0$  gives

$$\lambda_{1,-} - \lambda_1^+ = \frac{2\delta(\omega - r_1)}{nr_1} \frac{1}{1 + \delta/(1 + \omega)} + O(|\delta|n^{-4}),$$

which expands to (6.2). The proportional remainder follows because the secular equation is identical to the homogeneous one at  $\delta = 0$ ; tracking the estimates for the regular part above leaves a factor  $\delta$  in every omitted term.  $\blacksquare$

**Lemma 6.3** (*Right-edge root separation*). *Fix  $\omega > 0$  and  $\omega_0 \geq 0$ , and put  $D = (1 + \omega)/(2\omega)$ . There is  $N$  such that, for  $n \geq N$ , every nonzero eigenvalue other than the protected and affected roots born from  $q = \pm 2\pi/n$  satisfies*

$$\Re \lambda \leq \Re \lambda_1^+ - \pi^2 D n^{-2}. \quad (6.3)$$

The constants can be chosen uniformly when  $(\omega, \omega_0)$  ranges over a compact set with  $\omega$  bounded away from zero.

*Proof.* Put

$$\Omega_n = \{\lambda : \Re \lambda \geq \Re \lambda_1^+ - \pi^2 D n^{-2}\}.$$

The protected roots are explicit. Their slow branch satisfies, uniformly for small real  $q$ ,

$$\lambda^+(q) = -Dq^2 + O(q^4). \quad (6.4)$$

Choose  $q_* > 0$  so that the remainder is at most  $Dq^2/4$  for  $|q| \leq q_*$ . The modes with  $2\pi/n < |q| \leq q_*$  are then to the left of  $\Omega_n$ , while continuity puts both branches with  $q_* \leq |q| \leq \pi$  a fixed distance into the left half-plane. The fast small- $q$  branch is also separated from zero by a fixed amount. Thus, apart from zero, only the protected root born from  $q = \pm 2\pi/n$  lies in  $\Omega_n$  once  $n$  is large. We prove the same exclusion for the affected sector.

If  $\delta = 0$ , that sector is homogeneous and the conclusion follows from (6.4). Assume  $\delta \neq 0$  and set

$$f_n(\lambda) = 1 + 2\delta G_n(\lambda), \quad g_0 = \frac{1}{2(1 + \omega)}, \quad M_0 = 1 + 2\delta g_0 = \frac{1 + \omega_0}{1 + \omega} > 0. \quad (6.5)$$

In a neighborhood of zero, every affected root is a zero of  $f_n$ . Indeed, the residue at every slow homogeneous pole is nonzero, so multiplication by  $H_{n,-}^{(0)}$  cancels the pole of  $f_n$  but does not leave a zero there.

We first exclude affected zeros that do not tend to zero. Gershgorin's theorem places every finite-ring eigenvalue in a fixed disk and in the closed left half-plane. If the desired exclusion failed, there would be  $\epsilon > 0$  and a sequence of affected zeros in  $\Omega_n$  with  $|\lambda| \geq \epsilon$ ; a subsequence would converge to  $\lambda_* \neq 0$  with  $\Re \lambda_* = 0$ . The bulk spectral set meets the imaginary axis only at zero, so (5.9) gives  $G_n \rightarrow g_\infty$  locally uniformly at  $\lambda_*$ . The limiting secular equation would produce an  $\ell^2$  eigenvector  $u$  of the infinite defective generator with eigenvalue  $\lambda_*$ . This is impossible because

$$-\Re \langle u, L_\infty u \rangle = \frac{1}{2} \sum_{j,\sigma} |u_{j+1}^\sigma - u_j^\sigma|^2 + \sum_j \omega_j |u_j^+ - u_j^-|^2. \quad (6.6)$$

Equality in (6.6) forces an  $\ell^2$  vector to vanish. The subsequence contradiction therefore supplies  $\epsilon > 0$  such that no affected zero in  $\Omega_n$  can satisfy  $|\lambda| \geq \epsilon$ .

It remains to exhaust the zeros tending to zero. Put  $\lambda = \zeta/n^2$  and, using meromorphic continuation of (5.8), choose either local branch of

$$w_n(\zeta) = n \operatorname{arccosh} x(\zeta/n^2), \quad z(\zeta/n^2)^n = e^{-w_n(\zeta)}.$$

The quantities  $w_n^2$  and  $w_n \coth(w_n/2)$  below are even in  $w_n$ , so the choice does not affect the argument, including on the continued bulk cut. The definitions of  $a$  and  $x$  give

$$x - 1 = \frac{\omega}{1 + \omega} \lambda + O(\lambda^2), \quad \lambda + 1 - x = \frac{\lambda}{1 + \omega} + O(\lambda^2). \quad (6.7)$$

Since  $s = \sinh(w_n/n)$ , equations (5.8) and (6.7) give the uniform expansion on bounded  $\zeta$ -sets away from its poles

$$w_n(\zeta)^2 = \frac{\zeta}{D} + O(n^{-2}), \quad (6.8)$$

$$G_n(\zeta/n^2) = g_0 + \frac{w_n(\zeta)}{4\omega(1 + \omega)n} \coth\left(\frac{w_n(\zeta)}{2}\right) + O(n^{-2}). \quad (6.9)$$

The error in (6.9) remains  $O(n^{-2})$  when the distance from a nonzero pole of the hyperbolic cotangent is at least  $A/n$ , for fixed  $A > 0$ . The poles in the limiting  $\zeta$ -plane are  $-4\pi^2 D k^2$ . The scaled half-plane  $n^2 \Omega_n$  contains only the first one,  $-4\pi^2 D$ . Near that pole the second term in (6.9) is bounded by  $C/A$  outside  $|\lambda - \lambda_1^+| < An^{-3}$ . Choose  $A$  so that  $2|\delta|C/A < M_0/4$ . After increasing  $n$ , (6.9) gives

$$|f_n(\lambda)| \geq \frac{M_0}{2} \quad (6.10)$$

there, outside a disk  $|\lambda - \lambda_1^+| < An^{-3}$ , provided  $A$  and then  $n$  are large enough. If instead  $\lambda \in \Omega_n$  tends to zero while  $n^2|\lambda| \rightarrow \infty$ , then its imaginary part dominates its negative real part. Since  $x - 1 = \omega\lambda/(1 + \omega) + O(\lambda^2)$ , the standard square-root expansion of  $\operatorname{arccosh}$  gives  $\Re \operatorname{arccosh} x(\lambda) \geq c\sqrt{|\lambda|}$ . Consequently  $z(\lambda)^n \rightarrow 0$ , and (5.8) again gives  $G_n(\lambda) = g_0 + o(1)$  and (6.10). These two cases exhaust the region near zero.

Finally, on the circle  $|\lambda - \lambda_1^+| = An^{-3}$  write

$$G_n(\lambda) = \frac{2R_1}{n(\lambda - \lambda_1^+)} + G_n^{\text{reg}}(\lambda).$$

Equations (6.9) and (6.5) give  $1 + 2\delta G_n^{\text{reg}} = M_0 + O(n^{-1})$ . After multiplication by  $\lambda - \lambda_1^+$ , the secular equation becomes

$$(\lambda - \lambda_1^+)f_n(\lambda) = M_0(\lambda - \lambda_1^+) + \frac{4\delta R_1}{n} + O(n^{-1}|\lambda - \lambda_1^+|). \quad (6.11)$$

Here  $R_1 = O(n^{-2})$ . Enlarge  $A$ , if necessary, so that the zero of the first two terms lies strictly inside the circle. On the circle the last term is  $O(n^{-4})$ , whereas the first two terms have modulus bounded below by  $c_A n^{-3}$ . Rouché's theorem therefore counts exactly one

zero in the disk. It is the affected root of Lemma 6.2. Thus no other affected root occurs in  $\Omega_n$ , which proves the assertion.

All estimates above are uniform on the stated compact parameter sets:  $D$ ,  $\delta$ , and the generator norms are bounded there, while  $M_0$  has a positive minimum. ■

**Theorem 6.4** (*Fixed-rate gap asymptotics*). Fix  $\omega > 0$  and  $\omega_0 \geq 0$ .

1. If  $\omega_0 \leq \omega$ , then for every sufficiently large  $n$ ,

$$\gamma_n(\omega, \omega_0) = \gamma_n^{\text{hom}}(\omega). \quad (6.12)$$

2. If  $\omega_0 > \omega$ , then

$$\gamma_n(\omega, \omega_0) = \gamma_n^{\text{hom}}(\omega) - \frac{4\pi^2(\omega_0 - \omega)(1 + \omega)}{\omega^2(1 + \omega_0)} n^{-3} + O(|\omega_0 - \omega|n^{-4}). \quad (6.13)$$

3. In both cases,

$$\begin{aligned} \gamma_n^{\text{hom}}(\omega) &= \frac{2\pi^2(1 + \omega)}{\omega} n^{-2} \\ &\quad - 16\pi^4 \left( \frac{1}{24} + \frac{1}{6\omega} - \frac{1}{8\omega^3} \right) n^{-4} + O(n^{-6}). \end{aligned} \quad (6.14)$$

The errors are uniform when  $(\omega, \omega_0)$  ranges over a compact set with  $\omega$  bounded away from zero. The root selection is uniform through  $\omega_0 = \omega$ , where the two parity roots coincide.

*Proof.* Taylor expansion of (6.1) gives (6.14). Lemma 6.2 shows that, uniformly across  $\delta = 0$ , the affected first root moves toward zero precisely when  $\delta > 0$ . Lemma 6.3 shows that no other extended, band-edge, or localized root can compete with this first pair. Therefore the protected root wins when  $\delta \leq 0$ , proving (6.12), and the affected root wins when  $\delta > 0$ . Substitution of  $q^2 = 4\pi^2 n^{-2}$  in (6.2) then gives (6.13). ■

**Corollary 6.5** (*Fixed ratio, vanishing ratio, and a slow-site scaling*). Let the bulk rate  $\omega > 0$  be fixed.

1. If  $\omega_0/\omega \rightarrow \rho \in (0, \infty)$ , equations (6.12)–(6.14) apply. For  $\rho < 1$  the protected branch is selected; for  $\rho > 1$  the affected branch and its  $n^{-3}$  term are selected. At the boundary  $\rho = 1$ , the protected root is selected whenever  $\omega_0 \leq \omega$  and the affected root whenever  $\omega_0 > \omega$ , with the expansion uniform across equality.
2. If  $\omega_0/\omega \rightarrow 0$ , then (6.12) holds for every sufficiently large  $n$ .
3. In particular, if  $n\omega_0 \rightarrow \beta$  with  $\beta \geq 0$ , then (6.12) still holds. A single slow tumbling site therefore has no separate  $\omega_0 \asymp n^{-1}$  gap crossover when the bulk rate is fixed.

*Proof.* In items 2 and 3 one has  $\omega_0 < \omega$  for every sufficiently large  $n$ , so Theorem 6.4 applies. Item 1 follows by the same sign test and compact-uniform expansion. ■

**Lemma 6.6** (*Critical right-edge exhaustion*). Set  $\omega = \alpha/n$  and  $\omega_0 = \beta/n$ , where  $\alpha > 0$  and  $\beta \geq 0$  are fixed, and put  $k = 2\pi$ . For all sufficiently large  $n$ , every nonzero eigenvalue with maximal real part belongs to the protected or affected cluster born from  $q = \pm k/n$ . More precisely, if  $\Lambda_{1,n}$  is the largest real part within that cluster, then every eigenvalue outside the cluster satisfies

$$\Re \lambda \leq \Lambda_{1,n} - \begin{cases} cn^{-2}, & \alpha \leq k, \\ cn^{-1}, & \alpha > k, \end{cases} \quad (6.15)$$

for a constant  $c = c(\alpha, \beta) > 0$ .

*Proof.* The protected roots are the homogeneous roots, so only the affected sector requires an exhaustion argument. If  $\beta = \alpha$ , the two parity restrictions are homogeneous, the result follows directly from their explicit dispersion, and (6.20) below holds with zero shift. Assume henceforth that  $b := \beta - \alpha \neq 0$ . Write

$$p_{\ell,n}^{\pm} = \cos(\ell k/n) - 1 - \frac{\alpha}{n} \pm \sqrt{\frac{\alpha^2}{n^2} - \sin^2(\ell k/n)}$$

for its homogeneous poles. Away from a double pole, the paired  $q = \pm \ell k/n$  contribution to  $G_n$  has the partial-fraction form

$$\frac{2R_{\ell,n}^{\pm}}{n(\lambda - p_{\ell,n}^{\pm})}, \quad R_{\ell,n}^{\pm} = \frac{p_{\ell,n}^{\pm} + 1 - \cos(\ell k/n)}{p_{\ell,n}^{\pm} - p_{\ell,n}^{\mp}}. \quad (6.16)$$

We first prove an off-pole estimate that exhausts the affected roots. Gershgorin's theorem confines all eigenvalues to a fixed disk, so below we restrict the right-edge windows to that disk. If  $\alpha \leq k$ , choose  $B > 0$  large enough to contain the protected first root and, when  $\alpha < k$ , its affected  $O(n^{-2})$  displacement scale. We take  $B$  midway between two consecutive fixed-index coefficients  $\ell^2 k^2/2$  and consider

$$\Omega_{B,n} = \left\{ \Re \lambda \geq -\frac{\alpha}{n} - \frac{B}{n^2} \right\};$$

if  $\alpha > k$ , choose  $d$  strictly between

$$d_1 = \alpha - \sqrt{\alpha^2 - k^2}, \quad d_2 = \alpha - \Re \sqrt{\alpha^2 - 4k^2},$$

and put  $\Omega_{d,n} = \{\Re \lambda \geq -d/n\}$ . Only finitely many fixed-index pole centers lie in  $\Omega_{B,n}$ : indeed, their indices satisfy  $1 - \cos(\ell k/n) \leq Bn^{-2}$ . In  $\Omega_{d,n}$  only the slow first pole can occur for all sufficiently large  $n$ . Remove the disks of radius  $An^{-2}$  about these simple poles. When  $\alpha = k$ , replace the two first-pole disks by the single disk of radius  $An^{-3/2}$  about  $(p_{1,n}^+ + p_{1,n}^-)/2$ .

On the remaining set the exact Fourier sum satisfies

$$|G_n(\lambda)| \leq C_0(B) + C_1 \log n + C_2(B) \frac{n}{A} + C_3 \frac{n}{\sqrt{B}} \quad (\alpha \leq k), \quad (6.17)$$

where  $C_1, C_3$  are independent of  $A, B, n$  once  $B$  exceeds a fixed threshold. For  $\alpha > k$  the same bound holds without the  $C_3 n/\sqrt{B}$  term and with the remaining constants depending



on  $d$ . Here is the estimate. First combine the two roots belonging to each quadratic  $D_\ell$ ; this cancellation is essential. For  $|q_\ell| \leq q_0$ , the poles outside the finite set have horizontal distance at least  $c\ell^2 n^{-2} - Bn^{-2}$  from  $\Omega_{B,n}$ . After scaling imaginary parts by  $n$ , choose the pole whose scaled imaginary part is closest to  $n\Im\lambda$  and call its index  $\ell_0$ . Its contribution is at most  $Cn/\sqrt{B}$ ; the midpoint choice of  $B$  gives the required coefficient gap at the first exterior index. The  $j$ th neighboring index contributes at most  $C/(1+j)$ . This gives the last and logarithmic terms in (6.17). Each removed simple-pole term is at most  $Cn/A$  off its disk. At  $\alpha = k$ , the exact grouped first contribution, with  $\Delta = \lambda - (p_{1,n}^+ + p_{1,n}^-)/2$ , is

$$\frac{2(\Delta - k/n)}{n\{\Delta^2 - (p_{1,n}^+ - p_{1,n}^-)^2/4\}},$$

and is at most  $Cn/A^2$  outside the  $An^{-3/2}$  disk. Finally, the terms with  $|q_\ell| \geq q_0$  have denominators bounded away from zero in these right-edge windows and form a bounded Riemann sum. These four estimates prove (6.17).

Since  $2\delta = 2b/n$ , first enlarge  $B$  so that  $2|b|C_3/\sqrt{B} < 1/8$ , then enlarge  $A$  so that  $2|b|C_2(B)/A < 1/8$ , and finally increase  $n$  so that the constant and logarithmic terms contribute less than  $1/4$ . In the case  $\alpha > k$ , choose  $A$  first and then  $n$ . Thus (6.17) implies

$$|2\delta G_n(\lambda)| < \frac{1}{2}. \quad (6.18)$$

Thus every zero of  $1 + 2\delta G_n$  in a right-edge window lies in one of the displayed pole neighborhoods.

We also need the root count inside a simple-pole disk. There

$$G_n(\lambda) = \frac{2R_{\ell,n}^\pm}{n(\lambda - p_{\ell,n}^\pm)} + G_{\ell,n}^{\text{reg}}(\lambda), \quad G_{\ell,n}^{\text{reg}} = O(1).$$

The last bound follows directly from (4.3): for the remaining small wave numbers the combined quadratic summand is bounded by  $C/(1 + |j^2 - \ell^2|)$  up to  $j \asymp \sqrt{n}$ , while each of the remaining  $O(n)$  summands is  $O(n^{-1})$ . Multiplication of the secular equation by  $\lambda - p_{\ell,n}^\pm$  gives

$$(\lambda - p_{\ell,n}^\pm)(1 + 2\delta G_n) = (\lambda - p_{\ell,n}^\pm) \left( 1 + \frac{2b}{n} G_{\ell,n}^{\text{reg}} \right) + \frac{4bR_{\ell,n}^\pm}{n^2}. \quad (6.19)$$

Compare the right-hand side with

$$h_0(\lambda) = \lambda - p_{\ell,n}^\pm + 4bR_{\ell,n}^\pm n^{-2}.$$

Enlarge  $A$  once more so that the zero of  $h_0$  is strictly inside the disk. On its boundary, the difference from  $h_0$  is  $O(n^{-3})$ , whereas  $|h_0| \geq c_A n^{-2}$ . Rouché's theorem gives exactly one affected zero and

$$\hat{p}_{\ell,n}^\pm = p_{\ell,n}^\pm - 4bR_{\ell,n}^\pm n^{-2} + O(n^{-3}). \quad (6.20)$$

Together with (6.18), this accounts for every affected root in either right-edge window.

Suppose first that  $\alpha < k$ . For every fixed  $\ell \geq 1$ , the two poles are nonreal conjugates and

$$\Re R_{\ell,n}^{\pm} = \frac{1}{2}, \quad \Re p_{\ell,n}^{\pm} = -\frac{\alpha}{n} - \frac{\ell^2 k^2}{2n^2} + O(n^{-4}). \quad (6.21)$$

Equations (6.20)–(6.21) show that the same parity is selected for every fixed  $\ell$ , and that the  $\ell = 2$  cluster is to the left of the first by  $3k^2/(2n^2) + O(n^{-3})$ . The choice of  $B$  puts every possible competitor either among these finitely many clusters or to their left and separates all roots outside the window from the first cluster by a fixed multiple of  $n^{-2}$ . The endpoint tumble root has real part  $-2\alpha/n + O(n^{-2})$ . This proves (6.15) for  $\alpha < k$ .

At  $\alpha = k$ , the first protected cluster contains

$$-\frac{k}{n} + \left(-\frac{k^2}{2} + \frac{k^2}{\sqrt{3}}\right)n^{-2} + O(n^{-3}). \quad (6.22)$$

For every fixed  $\ell \geq 2$ , equations (6.20)–(6.21) apply with  $\alpha = k$ . If  $\beta \leq k$ , its largest possible real part is below (6.22) by

$$\left[\frac{(\ell^2 - 1)k^2}{2} + \frac{k^2}{\sqrt{3}} - 2(k - \beta)\right]n^{-2} + O(n^{-3}),$$

which is positive because  $k > 2\sqrt{3}$ . If  $\beta > k$ , the affected higher modes move to the left and the protected comparison is even stronger. Estimate (6.18) accounts for all remaining affected roots, proving the assertion at  $\alpha = k$ .

Finally, if  $\alpha > k$ , the first slow pole has decay coefficient  $d_1 = \alpha - \sqrt{\alpha^2 - k^2}$ . By the choice of  $d$ , the only protected pole in  $\Omega_{d,n}$  is the first slow pole, and (6.18)–(6.20) show that its affected partner is the only affected root there. Both have real part  $-d_1/n + O(n^{-2})$ , whereas every root outside the window has real part at most  $-d/n$ . This gives a separation of order  $n^{-1}$  and completes the proof. ■

**Theorem 6.7** (*Bulk-critical double scaling*). *Set*

$$\omega = \frac{\alpha}{n}, \quad \omega_0 = \frac{\beta}{n}, \quad k = 2\pi,$$

where  $\alpha > 0$  and  $\beta \geq 0$  are fixed. Away from  $\alpha = k$ ,

$$\alpha < k : \quad \gamma_n = \frac{\alpha}{n} + \left[\frac{k^2}{2} - 2(\alpha - \beta)_+\right]n^{-2} + O(n^{-3}), \quad (6.23)$$

$$\begin{aligned} \alpha > k : \quad \gamma_n &= \frac{\alpha - \sqrt{\alpha^2 - k^2}}{n} \\ &+ \left[\frac{k^2}{2} - 2(\beta - \alpha)_+ + \frac{\alpha - \sqrt{\alpha^2 - k^2}}{\sqrt{\alpha^2 - k^2}}\right]n^{-2} + O(n^{-3}). \end{aligned} \quad (6.24)$$

At  $\alpha = k$ ,

$$\beta \leq k : \quad \gamma_n = \frac{k}{n} + k^2 \left(\frac{1}{2} - \frac{1}{\sqrt{3}}\right)n^{-2} + O(n^{-3}), \quad (6.25)$$

$$\beta > k : \quad \gamma_n = \frac{k}{n} - 2\sqrt{k(\beta - k)}n^{-3/2} + O(n^{-2}). \quad (6.26)$$

Thus the first homogeneous exceptional scaling is a square-root boundary layer for a faster defect, while the side selecting the affected parity mode reverses as  $\alpha$  passes through  $2\pi$ .

*Proof.* For  $\alpha < k$ , the protected first pair has

$$\Re \lambda_1^+ = -\frac{\alpha}{n} - \frac{k^2}{2n^2} + O(n^{-3}).$$

The pole residue tends to  $R_1 = \frac{1}{2} + i\alpha/(2\sqrt{k^2 - \alpha^2})$ , and  $\delta = (\beta - \alpha)/n$ . Hence the affected real part shifts by  $-2(\beta - \alpha)n^{-2} + O(n^{-3})$  by (6.20). Taking the larger real part gives (6.23).

For  $\alpha > k$ , put  $s = \sqrt{\alpha^2 - k^2}$ . The protected rate is  $(\alpha - s)n^{-1} + k^2(2n^2)^{-1} + O(n^{-3})$ , while  $R_1 \rightarrow -(\alpha - s)/(2s)$ . The same pole equation shifts the affected eigenvalue by  $2(\beta - \alpha)(\alpha - s)/(sn^2) + O(n^{-3})$ . Selection of the larger real part gives (6.24). Lemma 6.6 shows in both cases that no root outside the first cluster can compete with the selected root.

At  $\alpha = k$ , the exact lattice splitting of the homogeneous first block is

$$\sqrt{\omega^2 - \sin^2(k/n)} = \frac{k^2}{\sqrt{3}n^2} + O(n^{-4}),$$

which proves (6.25) when the protected root wins. Near the center of this block, put

$$\lambda_c = \cos(k/n) - 1 - k/n, \quad r_n = \sqrt{(k/n)^2 - \sin^2(k/n)}, \quad \Delta = \lambda - \lambda_c.$$

The two  $q = \pm k/n$  terms can be retained exactly, while the remaining terms are uniformly bounded for  $|\Delta| \asymp n^{-3/2}$ :

$$G_n(\lambda) = \frac{2(\Delta - k/n)}{n(\Delta^2 - r_n^2)} + O(1) = -\frac{2k}{n^2\Delta^2} + \frac{2}{n\Delta} + O(1). \quad (6.27)$$

For completeness, the bounded remainder follows by pairing wave numbers  $\ell$  and  $n - \ell$ : for each fixed  $\ell \neq 1$  its magnitude is bounded by a constant times  $(\ell^2 - 1)^{-1}$ , the sum of these bounds converges, and the terms with  $|q_\ell|$  bounded away from zero form a bounded Riemann sum. For  $\beta > k$ , put  $b = \beta - k$ . Substitution of  $\Delta = n^{-3/2}y$  in  $1 + 2bG_n/n = 0$  gives  $y^2 = 4kb + O(n^{-1/2})$ . Rouché's theorem on disks of radius  $Cn^{-2}$  about the two approximations, with fixed sufficiently large  $C$ , gives two affected roots

$$\Delta = \pm 2\sqrt{kb}n^{-3/2} + O(n^{-2}).$$

The positive displacement gives (6.26). For  $\beta < k$ , put  $h = k - \beta > 0$  and substitute

$$\Delta = n^{-3/2}(y + xn^{-1/2})$$

in  $1 - 2hG_n/n = 0$ . The constant and  $n^{-1/2}$  coefficients in (6.27) give, respectively,

$$y^2 = -4kh, \quad x = 2h.$$

Rouché's theorem on two disks of radius  $Cn^{-5/2}$  about the resulting approximations, with fixed sufficiently large  $C$ , gives the affected pair

$$\Delta = \pm 2i\sqrt{kh}n^{-3/2} + 2hn^{-2} + O(n^{-5/2}). \quad (6.28)$$

Its decay rate exceeds the protected decay rate by

$$\left(\frac{k^2}{\sqrt{3}} - 2h\right)n^{-2} + O(n^{-5/2}).$$

This coefficient is positive for  $0 < h \leq k$  because  $k = 2\pi > 2\sqrt{3}$ . The protected root therefore wins for every  $\beta < k$ ; at  $\beta = k$  the two parity roots coincide. This proves (6.25); the case  $\beta > k$  was obtained from the positive displacement above. Lemma 6.6 supplies the required exclusion of every higher, endpoint, and off-pole affected root in both cases. ■

## 7. Telegraph comparison

**Definition 7.1** (Two continuum approximations). For a homogeneous ring of circumference  $n$ , the standard telegraph system is

$$\partial_t \rho = -\partial_x m, \quad \partial_t m = -\partial_x \rho - 2\omega m. \quad (7.1)$$

Its slow Fourier eigenvalue at wave number  $q$  is

$$\lambda_{\text{tel}}^+(q) = -\omega + \sqrt{\omega^2 - q^2}. \quad (7.2)$$

The lattice-diffusion-corrected system is

$$\partial_t \rho = -\partial_x m + \frac{1}{2}\partial_x^2 \rho, \quad \partial_t m = -\partial_x \rho + \frac{1}{2}\partial_x^2 m - 2\omega m. \quad (7.3)$$

Its slow eigenvalue is

$$\lambda_{\text{td}}^+(q) = -\frac{q^2}{2} - \omega + \sqrt{\omega^2 - q^2}. \quad (7.4)$$

**Theorem 7.2** (Exact relaxation rate versus telegraph rates). Fix  $\omega > 0$ , let  $q = 2\pi/n$ , and define  $\gamma_{\text{tel}} = -\lambda_{\text{tel}}^+(q)$  and  $\gamma_{\text{td}} = -\lambda_{\text{td}}^+(q)$ . Then

$$\gamma_n^{\text{hom}} - \gamma_{\text{tel}} = \frac{q^2}{2} - \left(\frac{1}{24} + \frac{1}{6\omega}\right)q^4 + O(q^6), \quad (7.5)$$

$$\gamma_n^{\text{hom}} - \gamma_{\text{td}} = -\left(\frac{1}{24} + \frac{1}{6\omega}\right)q^4 + O(q^6). \quad (7.6)$$

Consequently

$$\frac{\gamma_{\text{tel}}}{\gamma_n^{\text{hom}}} \longrightarrow \frac{1}{1+\omega}, \quad \frac{\gamma_{\text{td}}}{\gamma_n^{\text{hom}}} \longrightarrow 1. \quad (7.7)$$

In the bulk-critical scaling  $\omega = \alpha/n$ , the standard telegraph rate does recover the leading term:

$$\gamma_{\text{tel}} = \frac{\alpha - \Re\sqrt{\alpha^2 - (2\pi)^2}}{n}, \quad (7.8)$$

which agrees with the leading terms of (6.23)–(6.26).

*Proof.* The exact lattice dispersion replaces  $-q^2/2$  by  $\cos q - 1$  and  $q$  inside the square root by  $\sin q$ . Expanding  $\cos q$ ,  $\sin^2 q$ , and the square root gives (7.5)–(7.6). Dividing by the leading exact rate  $(1+\omega)q^2/(2\omega)$  gives (7.7). Substitution of  $\omega = \alpha/n$  and  $q = 2\pi/n$  into (7.2) gives (7.8). ■

**Proposition 7.3** (*Point-defect telegraph secular equation*). *On the circle cut at zero, define the point-defect telegraph operator by*

$$\mathcal{A}_\delta(\rho, m) = (-m', -\rho' - 2\omega m) \quad \text{on } (0, n),$$

with domain consisting of  $m, \rho \in H^1(0, n)$  having traces that satisfy

$$m(0+) = m(n-), \quad \rho(0+) - \rho(n-) = -2\delta m(0). \quad (7.9)$$

This is the distributional realization obtained by adding  $-2\delta \delta_0(x)m$  to the second equation in (7.1). Outside the homogeneous telegraph spectrum, put

$$\kappa = \sqrt{\lambda(\lambda + 2\omega)}$$

and interpret  $\kappa^{-1} \coth(n\kappa/2)$  as the even meromorphic function of  $\kappa$  obtained by analytic continuation; no sign choice for the square root is required. The affected-parity eigenvalues satisfy

$$1 + \delta \frac{\lambda}{\kappa} \coth\left(\frac{n\kappa}{2}\right) = 0. \quad (7.10)$$

The reflection-even telegraph modes are unchanged. Formula (7.10) is the continuum counterpart of (3.8); the exact lattice formula retains  $1 - \cos q$  and  $\sin q$ , and (7.5)–(7.6) quantify the resulting relaxation-rate error.

*Proof.* The jump in (7.9) cancels the delta distribution in  $-\rho' - 2\delta \delta_0 m(0)$ , while continuity of  $m$  prevents a delta distribution in  $-m'$ . Thus the stated domain gives the point interaction on  $L^2((0, n); \mathbb{C}^2)$ . It is densely defined because it contains  $C_c^\infty((0, n); \mathbb{C}^2)$ , and it is closed: the graph norm controls  $\|m'\|_2$  directly and controls  $\|\rho'\|_2$  by  $\|\rho'\|_2 \leq \|(\mathcal{A}_\delta(\rho, m))_2\|_2 + 2\omega\|m\|_2$ ; the trace theorem then preserves (7.9) under graph-norm limits.

Let  $\lambda$  lie outside the homogeneous telegraph spectrum. The eigenvalue equations away from zero are

$$-m' = \lambda \rho, \quad -\rho' - 2\omega m = \lambda m.$$

Thus  $\lambda \neq 0$  and elimination of  $\rho$  gives  $m'' = \kappa^2 m$ . Write

$$m(x) = A \cosh(\kappa(x - n/2)) + B \sinh(\kappa(x - n/2)).$$

The equality  $m(0) = m(n)$  gives  $B = 0$ , since  $\sinh(n\kappa/2) = 0$  would put  $\lambda$  in the homogeneous spectrum. Using  $\rho = -m'/\lambda$ , the second jump condition in (7.9) is equivalent to

$$m'(0) - m'(n) = 2\delta \lambda m(0). \quad (7.11)$$

Substitution of the even hyperbolic-cosine solution into (7.11) gives

$$-2\kappa \sinh(n\kappa/2) = 2\delta \lambda \cosh(n\kappa/2),$$

which is exactly (7.10).

Equivalently, the polarization entry of the homogeneous telegraph resolvent is

$$\frac{1}{n} \sum_{\ell \in \mathbb{Z}} \frac{\lambda}{\lambda(\lambda + 2\omega) + (2\pi\ell/n)^2} = \frac{\lambda}{2\kappa} \coth\left(\frac{n\kappa}{2}\right).$$

The series converges normally away from the homogeneous spectrum and the last expression supplies its meromorphic continuation when  $\kappa$  is purely imaginary, in agreement with the direct jump calculation. Finally, reflection-even modes have  $m(0) = 0$ , so the jump vanishes and those modes are protected exactly as in Lemma 4.2. ■

## 8. Conclusions

*Remark 8.1* (The theorem package). For the generator in Definition 2.1:

1. Theorem 3.3 and Theorem 3.4 give exact characteristic equations for every  $n, \omega$ , and  $\omega_0$ ; the trace equation is used away from its denominators and its cleared polynomial identity is used at their zeros.
2. Theorems 4.7 and 4.8 classify the algebraic and geometric multiplicities of every eigenvalue, including every exceptional point.
3. Theorem 5.2 gives an exact criterion and decay length for a defect-localized mode.
4. Theorems 6.4 and 6.7 give the fixed-rate, vanishing-defect, and bulk-critical large- $n$  gap asymptotics.
5. Theorem 7.2 and Proposition 7.3 compare the exact relaxation rate with the standard and diffusion-corrected telegraph approximations.

Each item retains the hypotheses stated in its cited theorem: in particular, localization is asserted off the bulk spectrum (with the compact case treated separately), finite-ring localization convergence assumes a simple noncompact root, and the gap results use their stated fixed-rate or double-scaling regimes.

## AI-use disclosure

The proof architecture and manuscript were developed with assistance from OpenAI GPT-5.6 Sol (`gpt-5.6-sol`) at max reasoning effort. The model served as an interactive research assistant for deriving and testing arguments, checking citations and calculations, writing exact verification code, and editing the exposition. The author set the research direction, evaluated and approved the final arguments, and assumes responsibility for the mathematical claims, references, computations, and text.

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